Reg. No. | | | | | | | | | | | [17EC45]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE–2019

ANALOG COMMUNICATION

Time: 3 Hours **Maximum Marks: 70**

Instructions:

i. **Answer any FIVE full Questions by choosing at least TWO questions from each part.**

ii. **Missing data may be suitably assumed.**

PART - A

1. a. State and prove the time scaling property and frequency shifting properties of Fourier Transform. **10**

   b. Evaluate the fourier transform of the signal $g(t)=\exp(-t).\text{Sin}2\pi f_c t.\ u(t)$. **04**

2. a. A sinusoidal carrier has amplitude of 10V and frequency of 30kHz. It is Amplitude Modulated by a sinusoidal voltage of amplitude 3V and frequency 1 kHz. **08**

   i. Determine equation for AM.

   ii. Determine modulation index.

   iii. Find total power when R=100ohms.

   iv. Obtain the frequency spectrum of the AM wave.

   b. Derive and explain the demodulation process involved in envelope detector with circuit diagram, equations and waveforms. **06**

3. a. Explain how the ring modulator is used in generation of DSBSC signals using necessary diagrams. **08**

   b. Evaluate how Costas Loop eliminates Phase synchronization problems in coherent detection. **06**

4. a. Explain the phase discrimination method of generating SSB signal. **08**

   b. Evaluate the performance of different amplitude modulation techniques with respect to bandwidth, power and transmission efficiency. **04**

   c. List the applications of different amplitude modulation techniques. **02**

PART - B

5. a. An Armstrong FM generator is used for the generation of WBFM with Δf = 75KHz and fc = 100MHz using narrow band carrier as 100KHz and second carrier as 9.5 MHz. Identify suitable multiplication factors. Assume m(t) range 100Hz to 15kHz. **08**

   b. Model Narrowband FM with necessary equations, phasor and block diagram. **06**

JAIN (Deemed to be University) Learning Resource Centre Jain Global Campus Ramanagar-112

6. a. Explain the generation of FM using direct method. **06**

b. Develop model of Phase Locked Loop(PLL) for demodulation of FM. **08**

7. a. Make use of Noise Factor in Cascade connection of two port Networks and develop Friss Formula. **08**

b. Prove that for DSBSC receiver with coherent detector the figure of merit is unity. **06**

8. Explain with block diagram the super heterodyne AM receiver and brief the design features of subsystems such as AF amplifier, mixer, local oscillator, detector and IF amplifier. **14**